Lecture 1: Probability Space

Instructor: Yichi Zhang

STAT-S520 Introduction to Statistics

Outline

Axioms of Probability

Finite Sample Spaces

Conditional Probability

Independence

Outline

Axioms of Probability

Finite Sample Spaces

Conditional Probability

Independence

Next Monday (Sep 8th)

- **Quiz 1:**
  - The first 20mins of the class.
  - Closed-book.
  - Two questions: one from Lecture 0 and one from today's lecture (among the first two topics).
  - Similar to the practice problems on Canvas.
- **HW 1:** Due in the midnight.

Probability?

- A long run frequency of occurrence.
  - If I toss a coin many many times, what's the proportion of heads?
  - If I ask GPT many many questions, what's the proportion of questions it could get it right?
- Your belief of the chances.
  - If I buy a lottery, what is my chance to win?

Axioms of Probability

Probability: Define the probability in life. Image you run an experiment and then observe an outcome (like tossing a coin)
You actually define a probability space. The probability space has some ingredients:

- The sample space S: A set of all possible outcomes of a certain experiment (the universe).
- An event E is a subset of outcomes in S. We say an event E happens if our observed outcome belongs to E.

Example

Consider tossing, and we get either head (H) or tail (T). What is S? Let the event be "I observe a head". What is the corresponding event?

Example

How about we toss two coins?

Axioms of Probability

Probability: Define the probability in life. Image you run an experiment and then observe an outcome (like tossing a coin)
You actually define a probability space. The probability space has some ingredients:

- The sample space S: A set of all possible outcomes of a certain experiment (the universe).
- An event E is a subset of outcomes in S. We say an event E happens if our observed outcome belongs to E.
- The set of all events $\mathcal{C}$: A collection of subsets (a collection of different Es) that we are interested about. We focus on the case that $\mathcal{C}$ includes all possible subsets of outcomes in this course.
- The probability measure P: A function that assigns the "probability" of an event (measure the chance of an event).

Axioms of Probability

Probability function P: Sometimes, we just know it.

Example

Image that you toss a **fair** coin twice.

$$P(\{HH\}) =$$

$$P(\text{first coin is head}) =$$

$$P(\text{I have at least one heads}) =$$

Axioms of Probability

Probability function P: Sometimes, we just know it.

Example

Image that you toss a **fair** coin twice.

$$P(\{HH\}) = \frac{1}{4}$$

$$P(\text{first coin is head}) = P(\{HT, HH\}) = \frac{2}{4} = \frac{1}{2}$$

$$P(\text{I have at least one heads}) = P(\{HT, HH, TH\}) = \frac{3}{4}$$

Axiom of Probability

Probability measure P

- The probability measure P is a function $P : \mathcal{C} \to [0, 1]$, such that
  - If $E \in \mathcal{C}$, then $0 \leq P(E) \leq 1$;
  - $P(S) = 1$;
  - If $A, B \subseteq \mathcal{C}$ and $A \cap B = \varnothing$ then

$$P(A \cup B) = P(A) + P(B).$$

- So the probability measure can also be viewed a *function* that assigns "masses" (with numbers between 0 and 1) to every event subset

Axiom of Probability

- The third rule is referred to as **countable additivity**
- Two-event countable additivity immediately implies many-event additivity:

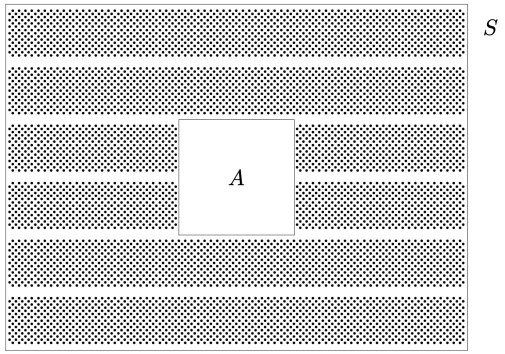
$$P\left(A_1 \cup A_2 \cup A_3 \cup \cdots\right) = P(A_1) + P(A_2) + P(A_3) + \cdots ,$$

when $A_i \cap A_j = \varnothing$ for any $i \neq j$.

Axiom of Probability

- If A is an event then $P(A^c) = 1 - P(A)$
- If A, B are events and $A \subset B$, then $P(A) \leq P(B)$
- If A, B are events then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Axiom of Probability



- $A \cap A^c = \varnothing$ and $A \cup A^c = S$; Therefore additivity implies

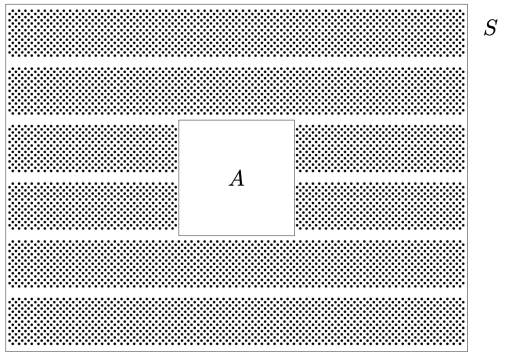
$$P(A) + P(A^c) = P(S) = 1 \quad \Longrightarrow \quad P(A^c) = 1 - P(A)$$

Example

Image that you toss a **fair** coin twice. $A = \{HH, HT\}$. Then $A^c =?$ and

$$P(A) + P(A^c) =?$$

Axiom of Probability



- $A \cap A^c = \varnothing$ and $A \cup A^c = S$; Therefore additivity implies

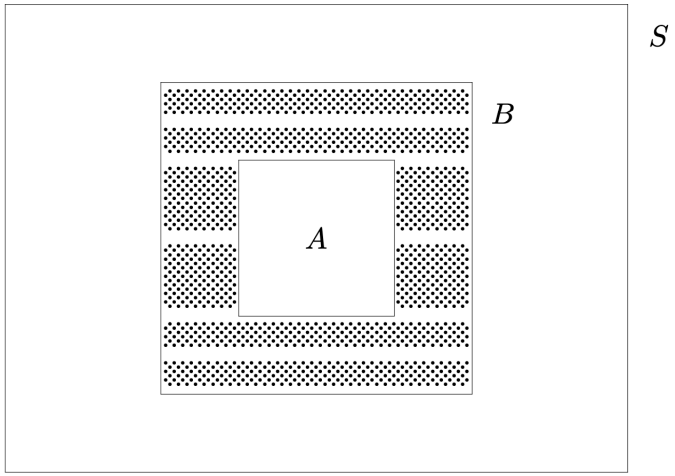
$$P(A) + P(A^c) = P(S) = 1 \quad \Longrightarrow \quad P(A^c) = 1 - P(A)$$

Example

Image that you toss a **fair** coin twice. $A = \{HH, HT\}$. Then $A^c = \{TH, TT\}$ and

$$P(A) + P(A^c) = \frac{2}{4} + \frac{2}{4} = 1$$

Axiom of Probability



- If A, B are events and $A \subset B$, then $P(A) \leq P(B)$
- Intuitively, $\mathrm{Mass}(B) \ = \mathrm{Mass}(\text{Shaded band}) + \mathrm{Mass}(A) \geq \mathrm{Mass}(A)$
- $A \cap (B \cap A^c) = \varnothing$ and $A \cup (B \cap A^c) = B$; Therefore

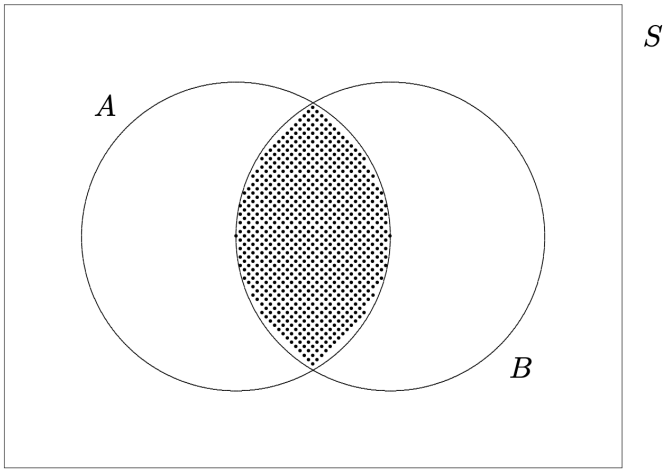
$$P(A) \leq \underbrace{P(A) + P((B \cap A^c)) = P(B)}_{\text{additivity}}$$

Example

Image that you toss a coin twice:

$$P(\text{the first coin is H}) \geq P(\text{HH})$$

Axiom of Probability



- We do not need $A \cap B = \varnothing$. We always have

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

- Intuitively, $\mathrm{Mass}(A) + \mathrm{Mass}(B) \ =$
$\mathrm{Mass}(\text{The region within A or B}) + \mathrm{Mass}(\text{The shaded region})$

Outline

Axioms of Probability

Finite Sample Spaces

Conditional Probability

Independence

Finite Sample Spaces

- Consider the case of a finite sample space $S = \{s_1, \ldots, s_N\}$.
- Take the set of events $\mathcal{C}$ be the collection of all subsets of S
- Let $p_i = P(\{s_i\})$
- Then the probability measure can be defined by

$$P(A) = \sum_{s_i \in A} p_i.$$

  Additivity!
- We focus on the case of equal probability outcomes: $p_1 = \ldots = p_N = 1/N$. Computation of probability of an event A reduces to counting problem:

$$P(A) = \sum_{s_i \in A} p_i = \sum_{s_i \in A} \frac{1}{N} = \frac{|A|}{|S|}.$$

Finite Sample Spaces

Example

Yichi has a class of 35 students:

- 4 undergrad in Data Science
- 7 undergrad not in Data Science
- 5 grad in Data Science
- 19 grad not in Data Science

He selects one student uniformly at random.

Finite Sample Spaces

Example

4 UG/DS, 7 UG/Non-DS, 5 G/DS, 19 G/Non-DS

- Yichi selects one student uniformly at random. What's the probability they're an undergrad in Data Science?

$$P(UG \cap DS) = \frac{4 \text{ undergrad DS}}{35 \text{ students}} = \frac{4}{35} \approx 11\%$$

- Yichi selects one student at random. What's the probability they're *not* an undergrad in Data Science?

$$P([UG \cap DS]^c) = 1 - P(UG \cap DS) = 1 - \frac{4}{35} = \frac{31}{35}$$

Finite Sample Spaces

Example

4 UG/DS, 7 UG/Non-DS, 5 G/DS, 19 G/Non-DS

- Yichi selects one student uniformly at random. What's the probability they're an undergrad?

$$P(UG) = \frac{4 + 7 = 11 \text{ undergrads}}{35 \text{ students}} = \frac{11}{35}$$

- Yichi selects one student uniformly at random. What's the probability they're in Data Science?

$$P(DS) = \frac{4 + 5 = 9 \text{ DS}}{35 \text{ students}} = \frac{9}{35}$$

Finite Sample Spaces

Example

4 UG/DS, 7 UG/Non-DS, 5 G/DS, 19 G/Non-DS

- Yichi selects one student uniformly at random. What's the probability they're an undergrad or in Data Science? (Note: This includes undergrad in Data Science.)

$$\mathrm{UG} \cup \mathrm{DS} = ?$$

$$\frac{4 \text{ UG DS} + 7 \text{ UG Non-DS} + 5 \text{ grad DS}}{35 \text{ students}} = \frac{16}{35}$$

Axiom of Probability

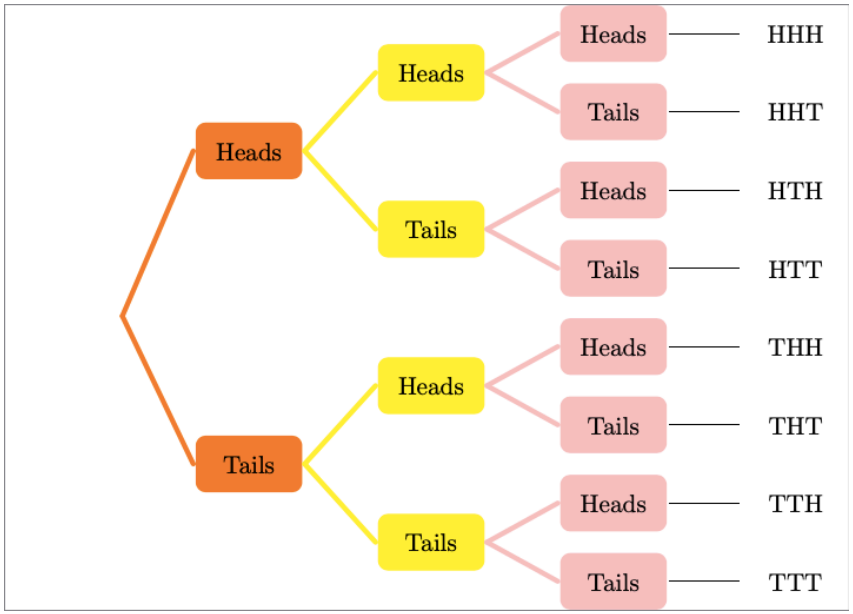
Example

Yichi tosses 3 fair coins. What is the probability he gets exactly two heads?

Axiom of Probability

Example

Yichi tosses 3 fair coins. What is the probability he gets exactly two heads?



Axiom of Probability

Example

- The probability of two heads is

$$\begin{aligned} P(2 \text{ heads}) &= \frac{\text{Number of ordered sequences with 2 heads}}{\text{Total number of ordered sequences}} \\ &= \frac{3}{8} \end{aligned}$$

- What is the probability of at least 1 head? ("At least 1" includes 1.) It's easier to count the number of outcomes that fall in the complement (i.e. zero heads), turn that into a probability, then subtract that from 1.

$$\begin{aligned} P(\text{at least one head}) &= 1 - P(\text{zero heads}) \\ &= 1 - \frac{1}{8} \\ &= \frac{7}{8}. \end{aligned}$$

Axiom of Probability

Example

- Yichi has a class of 35 students: 11 undergrads and 24 grad students.
- He selects two *different* students uniformly at random, *without* replacement, to form a committee.
- What is the probability both are undergrads?

Axiom of Probability

Example

- *Sample space:* There are 35 students, labeled as $\{x_1, \ldots, x_{35}\}$, and we choose 2 different students, so the sample space is

$$S = \{(x_i, x_j) : x_i \neq x_j, 1 \leq i, j \leq 35\}$$

- *Total number of committees:* cardinality of the entire sample space S is:

$$|S| = C(35, 2) = \frac{35!}{2!33!} = \frac{35 \times 34}{2 \times 1} = 595.$$

- *Number of possible committees with two undergrads:* There are 11 undergrad balls, and we want to choose 2, so this is:

$$C(11, 2) = \frac{11!}{2!9!} = \frac{11 \times 10}{2 \times 1} = 55.$$

$$P(\text{two undergrads}) = \frac{55}{595} = \frac{11}{119} \approx 9.2\%.$$

Outline

Axioms of Probability

Finite Sample Spaces

Conditional Probability

Independence

Conditional Probability

- Given events A and B with $P(B) > 0$, the conditional probability of A given B is defined to be

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- How to remember/understand: probability multiplication rule

$$\underbrace{P(A \cap B)}_{\text{events A and B both happen}} = \underbrace{P(B)}_{\text{event B happens}} \times \underbrace{P(A \mid B)}_{\text{event A happens given B happens}}$$

- Examples: $B = \{\text{I decide to do HW this morning}\}$ and $A = \{\text{I finish the HW this morning}\}$.

Conditional Probability

- **Law of total probability:** Let $(F_i)_{i=1}^n$ be a partition of the sample space S. Then for any event E,

$$P(E) = \sum_{i=1}^{n} P(E \mid F_i) P(F_i).$$

- Partition: several events such that one and only one of them will happen after the experiment.
- I have two questions and will randomly ask GPT one.

$$F_1 = \{\text{Ask GPT a math question}\},$$

$$F_2 = \{\text{Ask GPT a music question}\},$$

$$E = \{\text{The answer by GPT is right}\}.$$

- We can choose one question from n questions, then we have $F_1, \ldots, F_n$.

Conditional Probability

- **Bayes formula:** Let A and B be two events,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

TikTok viral videos

TikTok wants to understand how audience behaves to viral videos.
Setup. Suppose we know

$$P(V) = 0.1, \quad P(L \mid V) = 0.8, \quad P(L \mid V^c) = 0.2,$$

where $V = \{$A random video feed to the new user is viral$\}$,
$L = \{$The new user likes the randomly feed video$\}$.

Questions.

1. Find $P(L)$.
2. Find $P(V \mid L)$.

TikTok viral videos

TikTok wants to understand how audience behaves to viral videos.
Setup. Suppose we know

$$P(V) = 0.1, \quad P(L \mid V) = 0.8, \quad P(L \mid V^c) = 0.2,$$

where $V = \{$A random video feed to the new user is viral$\}$,
$L = \{$The new user likes the randomly feed video$\}$.

Questions.

1. Find $P(L)$.
2. Find $P(V \mid L)$.

Solution.

$$\begin{aligned} P(L) &= P(L \mid V)P(V) + P(L \mid V^c)P(V^c) \\ &= (0.8)(0.1) + (0.2)(0.9) = 0.26. \end{aligned}$$

$$P(V \mid L) = \frac{P(L \mid V)P(V)}{P(L)} = \frac{0.8 \times 0.1}{0.26} \approx 0.308.$$

Conditional Probability Function is also a Probability Function

- Given any B happened, conditional probability $P(\cdot \mid B)$ is also a probability measure.
- Properties of probability measure P also apply to $P(\cdot \mid B)$
- $P(A^c \mid B) = 1 - P(A \mid B)$ for any event A
- $P(A \mid B) \leq P(C \mid B)$ for any events A, C with $A \subset C$
- etc.

Independence

- Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B)$$

- More generally, a collection of events $A_1, \ldots, A_n$ are mutually independent if for **any** sub-collection of them, the multiplication rule holds. For example,

$$P(A_1 \cap A_3 \cap A_5) = P(A_1) \cdot P(A_3) \cdot P(A_5)$$

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

$$\ldots\ldots$$

Independence

- Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B)$$

- More generally, a collection of events $A_1, \ldots, A_n$ are mutually independent if for **any** sub-collection of them, the multiplication rule holds. For example,

$$P(A_1 \cap A_3 \cap A_5) = P(A_1) \cdot P(A_3) \cdot P(A_5)$$

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

$$\ldots\ldots$$

The formula can be used in two ways:

- Based on yours understanding of the experiment, assume that two events are independent and use the defining equation to compute the probability that both occur.
- Check independence of two (or more) events by calculating whether or not the definitions above are satisfied.

Independence

Example

Yichi rolls a fair six-sided die. Recall that digit "1" and digit "4" are square numbers. Are the events

- A: Yichi rolls an even number
- B: Yichi rolls an square number

independent?

Independence

Example

Yichi rolls a fair six-sided die. Recall that digit "1" and digit "4" are square numbers. Are the events

- A: Yichi rolls an even number
- B: Yichi rolls an square number

independent?

Solution: We need to check if

$$P(A \cap B) = P(A) \cdot P(B)$$

is true.

$$\begin{aligned} P(A) &= 3/6 = 1/2 \\ P(B) &= 2/6 = 1/3 \\ LHS &= P(A \cap B) = P(4) = 1/6 \\ RHS &= 1/2 \times 1/3 = 1/6 \end{aligned}$$

So independence holds.

Independence

If $P(A \mid B) \neq P(A)$ then A and B are *not* independent. Why? if they are independent, conditional on B will not affect the happening probability of A. Or mathematically we have, if A and B are independent, we must have

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

So $P(A \mid B) \neq P(A)$ means they are not independent.

Example

Approximately 85% of all U.S. residents who are struck by lightning are men. Consider the population of U.S. residents, from which a person is randomly selected. Let A denote the event that the person is male and let B denote the event that the person will be struck by lightning. Are A and B independent events?

Independence

If $P(A \mid B) \neq P(A)$ then A and B are *not* independent. Why? if they are independent, conditional on B will not affect the happening probability of A. Or mathematically we have, if A and B are independent, we must have

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

So $P(A \mid B) \neq P(A)$ means they are not independent.

Example

Approximately 85% of all U.S. residents who are struck by lightning are men. Consider the population of U.S. residents, from which a person is randomly selected. Let A denote the event that the person is male and let B denote the event that the person will be struck by lightning. Are A and B independent events?

Solution:

- We are told that $P(A \mid B)$ is approximately 0.85.
- We know that $P(A)$ is roughly 0.5.
- Because $P(A \mid B) \neq P(A)$, we conclude that A and B are dependent events.

Random Variables

Random variable is a function of experiment outcome $s \in S$

$$X(s) : S \to \mathbb{R}$$

$$\text{Experiment} \Rightarrow \text{Outcome } s \Rightarrow X = X(s)$$

Note: Later in this course we will often encounter multiple random variables defined on the same probability space (same experiment). For two random variables X_1 and X_2, we can further define new random variables:

- $(X_1 + X_2)(s) := X_1(s) + X_2(s)$;
- $(X_1 X_2)(s) := X_1(s) X_2(s)$;
- $(X_1 / X_2)(s) := X_1(s) / X_2(s)$;

Random Variables

$$\text{Experiment} \Rightarrow \text{Outcome } s \Rightarrow X = X(s)$$

Example

- Suppose that two coins are tossed and Y counts the number of Heads in each outcome:

$$Y(\mathsf{HH}) = 2, \quad Y(\mathsf{HT}) = 1,$$
$$Y(\mathsf{TH}) = 1, \quad Y(\mathsf{TT}) = 0.$$

- Suppose $X_1 = 1$ if the first coin is head and $X_1 = 0$ if the first coin is tail.
- Suppose $X_2 = 1$ if the second coin is head and $X_1 = 0$ if the second coin is tail.
- Are X_1 and X_2 functions? How do we represent them?
- Then you build random variable Y by X_1 and X_2: $Y = X_1 + X_2$

Random Variables

- **Distribution of the random variable X:** If we repeat the experiment many times and record the realizations of X, the resulting collection of values forms the distribution of X.
- You look at the distribution of these randomly generated $X_1, X_2, \ldots$ on the real line ($\mathbb{R}$), you have a sense about P_X.
- Example: X is a number among $0, 1, 2, 3, 4, 5$.

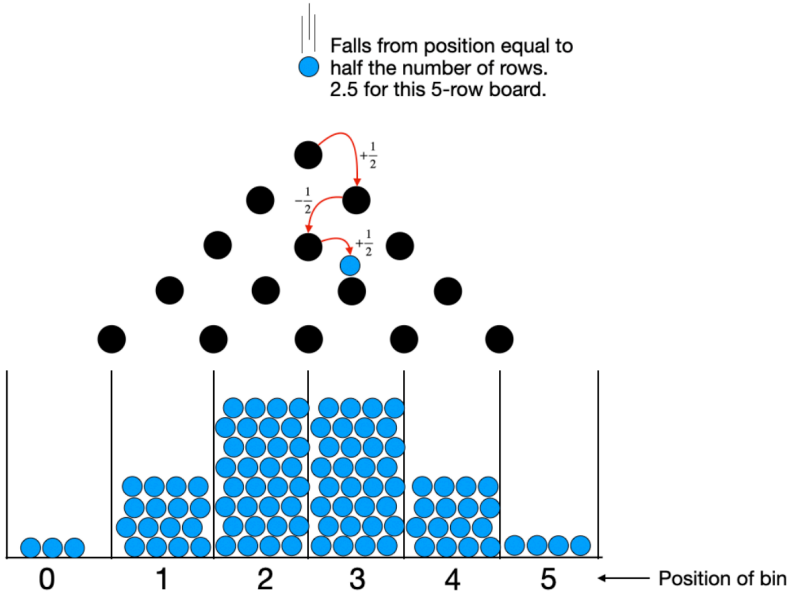


Fig. 1: Galton Board overview

Random Variables

- **Distribution of the random variable X:** If we repeat the experiment many times and record the realizations of X, the resulting collection of values forms the distribution of X.
- You look at the distribution of these randomly generated $X_1, X_2, \ldots$ on the real line ($\mathbb{R}$), you have a sense about P_X.
- The probability (likelihood) associated with the occurrence of X on each location of $\mathbb{R}$, characterizes the distribution.
- We use P_X to represent the distribution of X.

Random Variables

- How to represent the distribution of X in statistics (and math)?
- We use **cumulative distribution function** (CDF) of the random variable X:

$$F(x) = P(X \leq x) = P(\{s \in S : X(s) \leq x\})$$

  the probability that X assigns a real number not greater than x
- We know the value of $F(x)$ for all varying x, then we know the distribution of X. Why? let's see this in the following two slides!

Random Variables

- If $x_1 \leq x_2$, then

$$\{s \in S : X(s) \leq x_1\} \subset \{s \in S : X(s) \leq x_2\}$$

  which implies that F is a non-decreasing function:

$$F(x_1) \leq F(x_2).$$

- $F(-\infty) = 0$ and $F(+\infty) = 1$ for any cdf F.
- F uniquely defines a distribution on $\mathbb{R}$
- Given a cdf F, there exists a random variable X defined on a probability space s.t. $F(x) = P(X \leq x)$.

Random Variables

You can use the CDF to find the probability that a random X belongs to any reasonable sets on $\mathbb{R}$. For example:

- $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$
- $P(a < X \leq b) = P(X \leq x) - P(X \leq a) = F(b) - F(a)$
- $P(X = x) = P(X \leq x) - P(X < x)$
- $P(X \leq x)$ is just $F(x)$.
- $P(X < x)$ is the "limit from below" of $F(x)$ at x: Technically, it is given as the limit

$$P(X < x) = \lim_{y \to x-} P(X \leq y) = \lim_{y \to x-} F(y)$$

Random Variables

Let's plot CDF.

Example

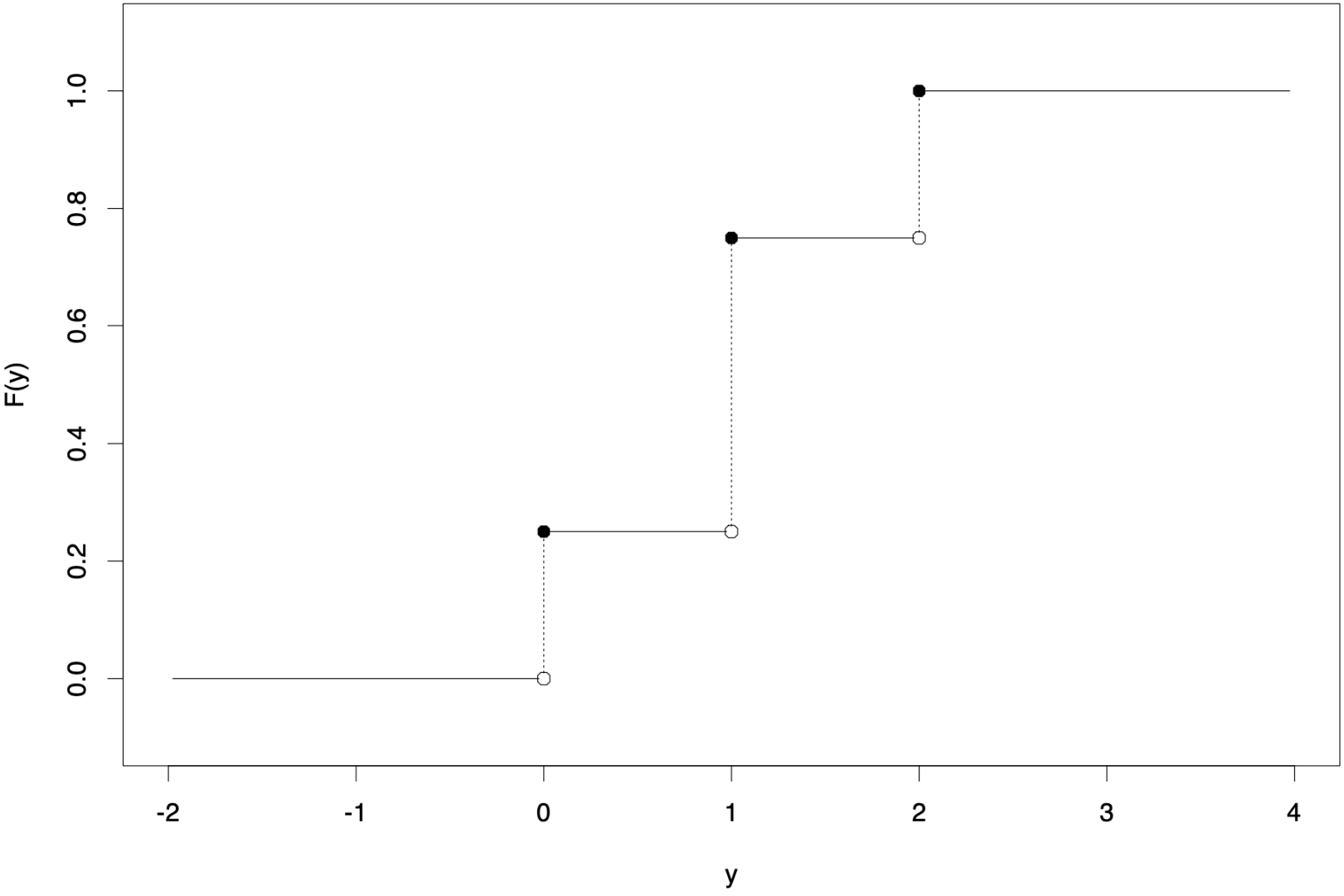
Suppose that two fair coins are tossed independently, so that each of the four possible outcomes are equally likely. Recall the random variable Y that counts the number of Heads. It follows that

$$P(Y = 0) = 1/4, \quad P(Y = 1) = 1/2, \quad P(Y = 2) = 1/4.$$

Then

- If $y < 0$, then $F(y) = P(Y \leq y) = P(\varnothing) = 0$ because $Y \geq 0$
- $F(0) = P(Y \leq 0) = P(Y = 0) = 1/4$
- If $y \in (0, 1)$ then $F(y) = P(Y \leq y) = P(Y = 0) = 1/4$
- $F(1) = P(Y \leq 1) = P(Y = 0) + P(Y = 1) = 3/4$
- If $y \in (1, 2)$, then $F(y) = P(Y \leq y) = P(Y = 0) + P(Y = 1) = 3/4$
- If $y \geq 2$, then $F(y) = P(Y \leq y) = P(Y \leq 2) = 1$

Random Variables



Random Variables

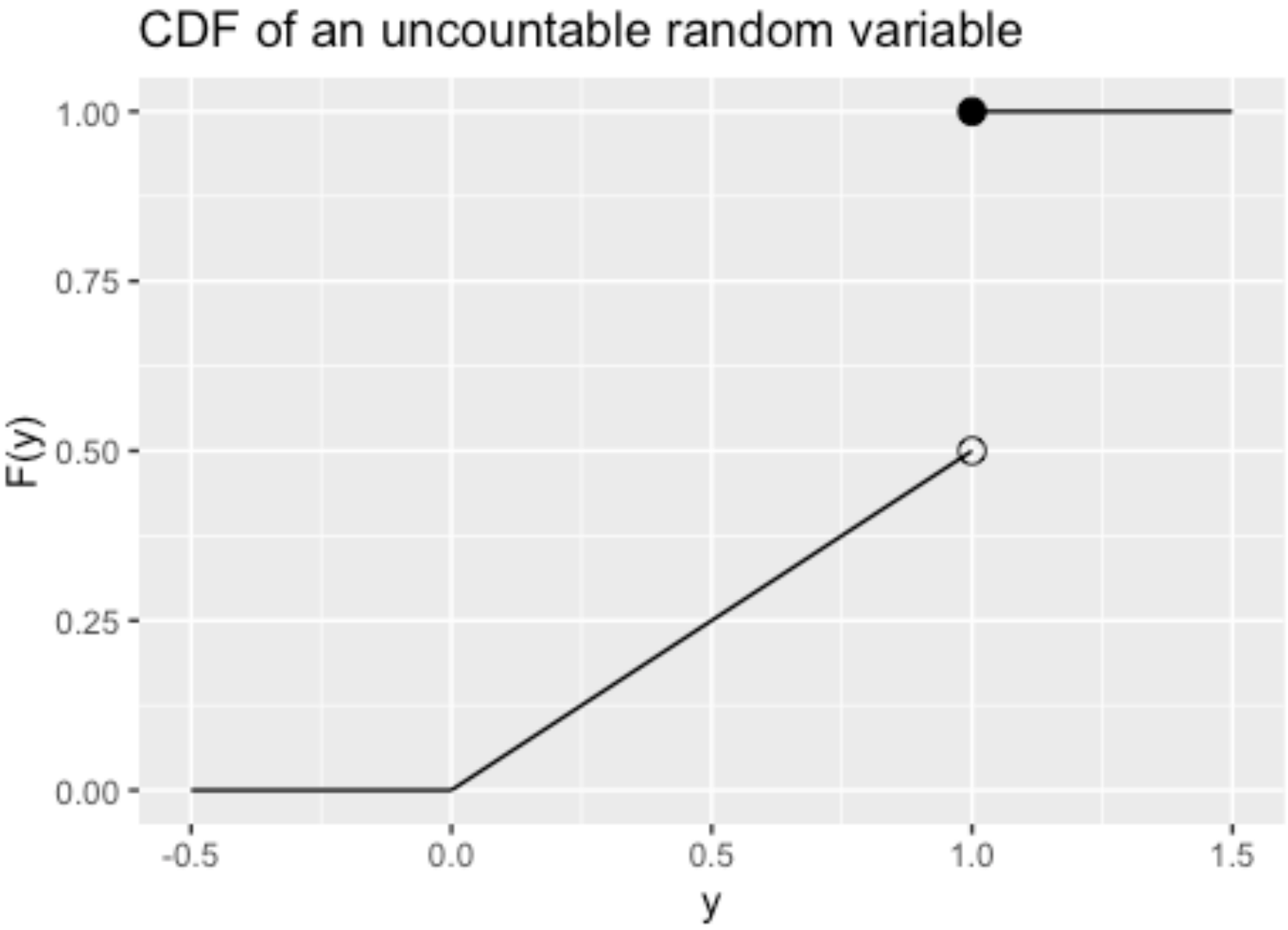
Example

Let the CDF of random variable X be

$$F(y) = \begin{cases} 0 & y < 0 \\ y/2 & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases} .$$

1. What's $P(X > 0.5)$?
2. What's $P(0.2 < X \leq 0.8)$?
3. What's $P(X = 1)$?

Random Variables



Random Variables

Example

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}.$$

1. *What's* $P(X > 0.5)$*?*

$$\begin{aligned} P(X > 0.5) &= 1 - P(X \leq 0.5) \\ &= 1 - F(0.5) \\ &= 1 - (0.5/2) \\ &= 0.75. \end{aligned}$$

Random Variables

Example

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}.$$

2. *What's* $P(0.2 < X \leq 0.8)$*?*

$$\begin{aligned} P(0.2 < X \leq 0.8) &= P(X \leq 0.8) - P(X \leq 0.2) \\ &= F(0.8) - F(0.2) \\ &= (0.8/2) - (0.2/2) \\ &= 0.3. \end{aligned}$$

Random Variables

Example

$$F(x) = \begin{cases} 0 & x < 0 \\ x/2 & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}.$$

3. What's $P(X = 1)$?

$$\begin{aligned} P(X = 1) &= P(X \leq 1) - P(X < 1) \\ &= F(1) - F(0.999\ldots) \\ &= 1 - (0.999\ldots/2) \\ &= 1 - 0.5 \\ &= 0.5. \end{aligned}$$

Random Variables

- Two random variables X_1 and X_2 defined on the same probability space is **independent** if for any $x_1, x_2 \in \mathbb{R}$,

$$P(X_1 \leq x_1, X_2 \leq x_2) = P(X_1 \leq x_1)P(X_2 \leq x_2)$$

- Let $X_1, \ldots, X_n$ be n random variables. Then we say $X_1, \ldots, X_n$ are independent, or mutually independent, if for any $x_1, \ldots, x_n \in \mathbb{R}$,

$$P(X_1 \leq x_1, \ldots, X_n \leq x_n) = P(X_1 \leq x_1) \ldots P(X_n \leq x_n)$$

Random Variables

- When we discuss random variables, we implicitly treat them as functions defined on the probability space
- But in the world of statistics we care more about their distributions
- So we often just write X but not $X(s)$. It could viewed as a random number generated after the experiment is done. But different random variables (random numbers), because they have different meanings (they are different functions), they will have different distributions. In the following course, we will study the distributions of different random variables carefully.